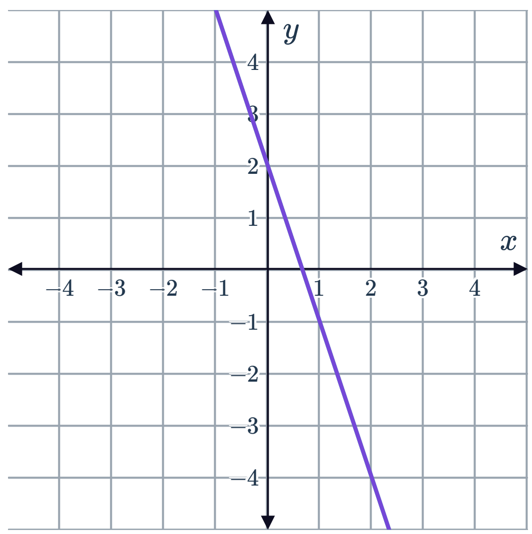
Graphing systems of equations

1 State whether the following graphs show a system of equations with:

- No solutions
- One solution
- Infinitely many solutions

a



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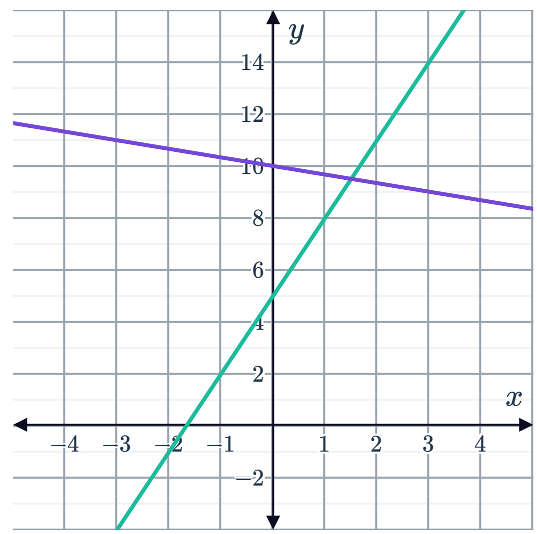

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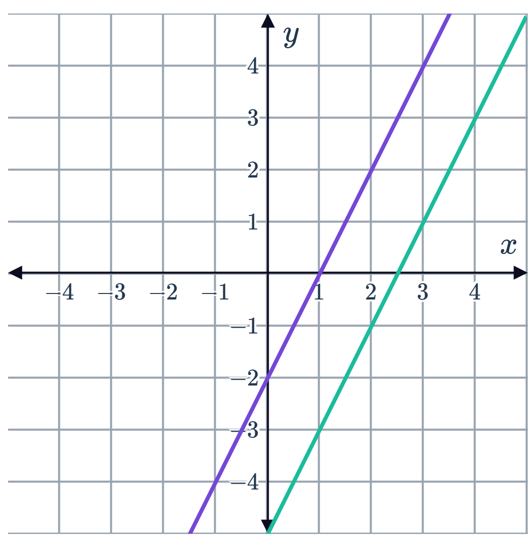

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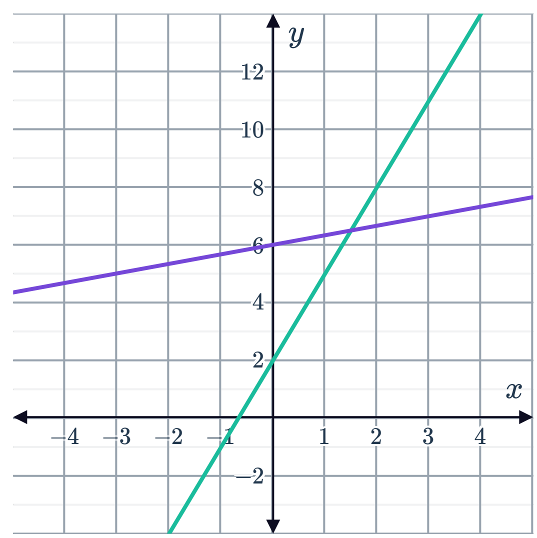

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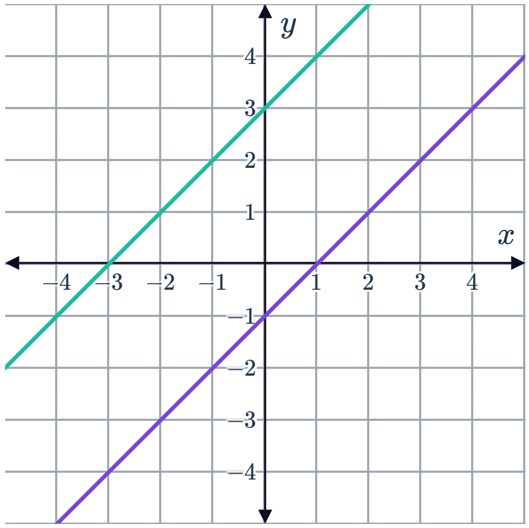
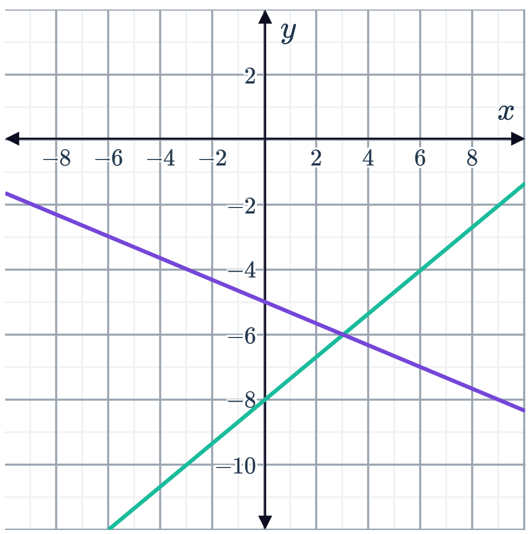


2 State the solution to the following systems of equations in the form (x, y).

a



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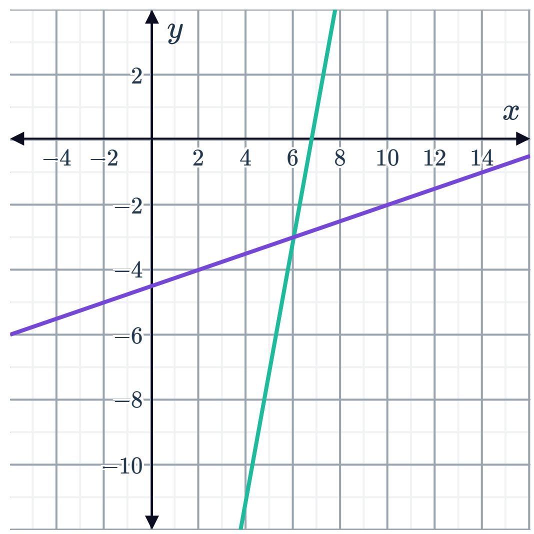

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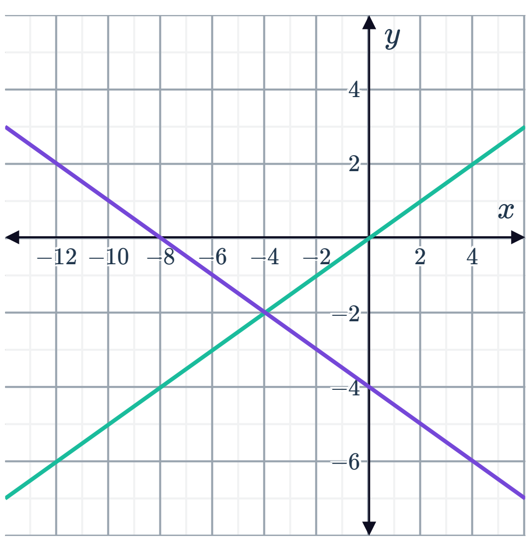


3 Consider the following linear equations:

- Equation 1: $y = 4x - 3$
- Equation 2: $y = 4 - 3x$

x	-3	-2	-1	0	1	2	3
y							

a Fill in the table of values using the line $y = 4x - 3$.

b Fill in the table of values using the line $y = 4 - 3x$.

c State the slope of the line $y = 4x - 3$.

d State the slope of the line $y = 4 - 3x$.

e Sketch the graph of both lines on the same coordinate axes.

4 Consider the following linear equations:

- $y = \dfrac{x}{3} + \dfrac{1}{3}$
- $-8y = 8x + 8$

a Determine the intercepts of the line $y = \dfrac{x}{3} + \dfrac{1}{3}$.

b Determine the intercepts of the line $-8y = 8x + 8$.

c Sketch the graph of both lines on the same coordinate axes.

d State the values of x and y which satisfy both equations.

5 Consider the following linear equations:

- $y = 2x + 2$
- $y = -2x + 2$

a Determine the slope and y-intercept of the line $y = 2x + 2$.

b Determine the intercepts of the line $y = -2x + 2$.

c Sketch the graph of both lines on the same coordinate axes.

d State the values of x and y which satisfy both equations.

6 Consider the following systems of linear eqns:

i Sketch the graph of both lines on the same coordinate axes.

ii State if there exists a value for x and y that satisfy the two equations simultaneously. If yes, state the values of x and y.

a $y = -4x - 1$
$y = -4x + 2$

b $4x - 2y = 2$
$-2x + 4y = 2$

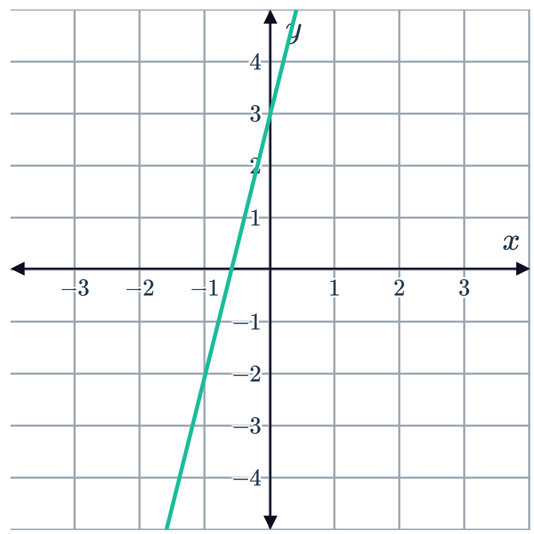
c $y = 5x - 7$
$y = -x + 5$

d $y = x + 0$
$y = -1$

7 A system of linear equations has no solutions. One of the equations of the system is $y = -4x - 3$. Which equation could be the other equation of the system?

- $y = -4x - 4$
- $y = -\dfrac{x}{4} - 3$
- $y = 4x + 3$
- $y = \dfrac{x}{4} - 4$

8 Consider the graph of the equation $y = 5x + 3$:
If a second line $y = mx + b$ intersects this line at the point $(0, 3)$, which of the following statements is true?



- $b = 3$
- $m > 5$
- $m = 5$
- $m < 0$

9 Consider the system of linear equations:

- Equation 1: $y = x - 2$
- Equation 2: $y = 5x - 6$

a Add equations 1 and 2 to create equation 3. State equation 3.

b Sketch the graph of equations 1, 2 and 3 on the same coordinate axes.

c Determine the solution to the system of equations.

10 Consider the system of linear equations:

- Equation 1: $2x + y = -2$
- Equation 2: $2x + 5y = 14$

a Multiply Equation 1 by 3 and add it to Equation 2 to create Equation 3. State Equation 3.

b Sketch the graph of equations 1, 2 and 3 on the same coordinate axes.

c Determine the solution to the system of equations.

11 A rectangular zone is to be $4\ \text{ft}$ longer than it is wide, with a total perimeter of $32\ \text{ft}$.
Let y represent the length of the rectangle and x represent the width.

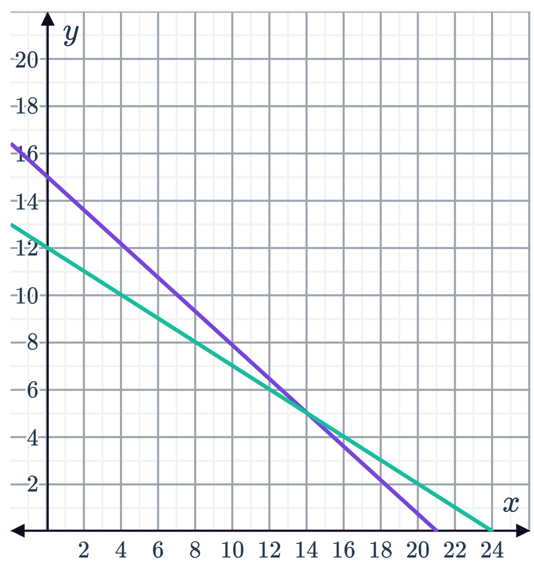
a Write the two equations that represent the information.

b Sketch the graph of the two equations on the same axes.

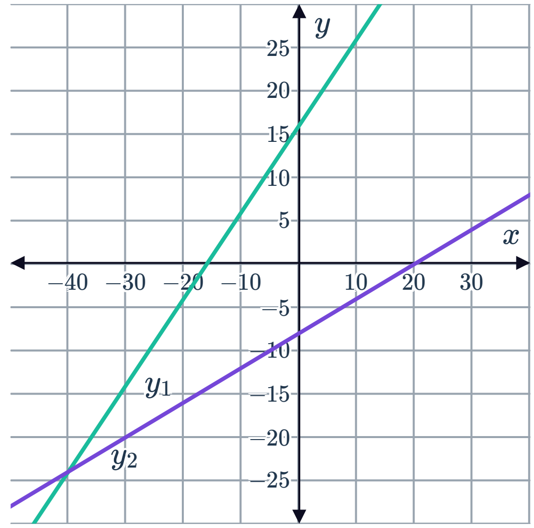
c Hence, find the values of the length and width of the rectangle.

Additional questions

12 Write a scenario to represent the system of equations and its solution. Explain what the solution to the system means in terms of the scenario.



13 Two equations, y_1 and y_2 represent the growth of two different house plants over time. Use the graph of y_1 and y_2 to support the claim that the two plants will never reach the same height on the same day.



14 Describe a situation where it would be unfeasible to solve a system of equations by graphing.